

An Agentic Retrieval-Augmented Generation System for EU Women's Safety Policy

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Abstract

We present an agentic retrieval-augmented generation (RAG) system that answers questions about women's safety law and policy in the European Union. The system combines a hybrid dense/sparse retriever over a 12,863-chunk corpus of 33 legal and research documents with a five-agent team coordinated by an LLM router, exposed through a Model Context Protocol (MCP) server providing eight tools for live web search, document ingestion, vector-store mutation, and artefact generation. A human-in-the-loop (HITL) review gate governs publication.

Our contribution beyond a conventional RAG baseline is threefold. First, we treat *every* text crossing the trust boundary — search snippets, scraped pages, user-supplied PDFs, and even passages retrieved from our own vector store — as potentially adversarial, and neutralise instruction-injection payloads in place rather than dropping the containing document. Second, we bound each agent's capability in code through a role-permission table, so a successful prompt injection cannot escalate to a tool the compromised agent was never entitled to call. Third, we instrument the entire pipeline with a hierarchical tracer that records the agent call tree, per-call token cost, and the exact prompt window at each LLM invocation, making the "thought process" auditable after the fact.

We evaluate retrieval and generation on a 20-question adversarial benchmark using an LLM-as-judge protocol and the RAGAS framework, and evaluate the security layer against a corpus of injection, exfiltration, SSRF, and path traversal payloads.

A central methodological finding is that the retrieval metric we began with was measuring the wrong quantity. Scoring retrieval by whether a clause of the reference answer appears verbatim in the retrieved text reports 33.3%; scoring the same retrievals by whether the passages actually contain the answer reports **60.0%**, and the two metrics agree on only 26.7% of questions. Local RAGAS corroborates the higher figure with a context recall of 0.854. Much of the apparent weakness of this system was an artefact of lexical scoring, and we report all three metrics rather than the most favourable one.

Generation pass rate is 46.7% on a benchmark deliberately weighted toward hard clause-level legal lookups (11 of 15 scored questions). The security layer blocks every tested attack payload across 253 automated tests. Total API expenditure across all development and evaluation was \$0.17.

1. Introduction

1.1 Motivation

Information about women's safety rights in the EU is distributed across instruments that are individually authoritative but collectively hard to navigate: the Council of Europe Istanbul Convention, Directive (EU) 2024/1385 on combating violence against women, the Victims' Rights Directive, the Digital Services Act, and a large grey literature of EIGE and FRA research reports. A person trying to establish what protection they are entitled to must reconcile overlapping instruments with different scopes, transposition deadlines, and enforcement mechanisms.

This is a natural RAG problem: the answer exists in documents, but finding and synthesising it requires retrieval plus reasoning. It is also a domain where the cost of a confident wrong answer is high, which shapes every design decision reported here.

1.2 Why agentic, and where agents are unsuitable

An agent adds value over plain RAG only when the task requires actions the retriever cannot perform: consulting sources outside the corpus, verifying whether a corpus claim has since been superseded, or producing artefacts. Our corpus has a fixed knowledge cutoff, and EU legislation moves; a fact-checking agent with live web access addresses exactly that gap.

Equally, agents are a poor fit for highly sensitive tasks, because agents fail in ways that are hard to detect: hallucination, confidently wrong answers, and blind trust in tool output. Legal advice affecting a person at risk is a sensitive task. Our response is not to avoid the agentic architecture but to constrain it: agents research and draft, a human approves, and the system is designed so that reviewing the agent's work is faster than doing the research manually. The observability layer exists precisely to make review fast.

1.3 Contributions

1. **A defence-in-depth security layer** (`app/security.py`) covering prompt injection, data leakage, and output validation, with SSRF and path-traversal protection for the tool surface. Sanitisation is in-place rather than rejecting, so a single injected sentence does not discard an otherwise useful source.
 2. **Capability bounding by role** (`_ROLE_PERMISSIONS` in `app/agents.py`), enforcing in code what a system prompt can only request.
 3. **A dependency-free hierarchical tracer** (`app/observability.py`) recording the agent tree, token cost per call, and the prompt window at each step.
 4. **An eight-tool MCP server** including Mermaid diagram generation alongside matplotlib charts, letting the Visualizer agent express structure as well as magnitude.
 5. **A 207-test offline suite** that runs with no API key, no network, and no prebuilt index, so correctness is verifiable on a clean checkout.
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2. Corpus Description

The corpus is a collection of EU and Council of Europe instruments and research reports on women's safety: the Istanbul Convention, Directive (EU) 2024/1385 on combating violence against women, the Victims' Rights Directive, the Digital Services Act, and grey literature from EIGE and FRA.

Property	Value
Source documents	33
Pages processed	2,247
Chunks indexed	12,863
Mean chunk length	670 characters
Chunk target / overlap	800 / 200 characters
Chunks carrying structured identifiers	3,811 (30%)
Index	FAISS IndexFlatIP , 12,863 × 768, L2-normalised
Sparse index	BM25 (Okapi)

The processing that produced it — extraction, cleaning, segmentation, and metadata management — is described below; the experiments that justify the segmentation parameters are in §4 and §5.

8.1 Text extraction

Extraction is dual-path. PyMuPDF (`fitz`) handles the bulk because it is fast and preserves reading order on well-formed PDFs. When a page yields fewer than `FALLBACK_CHAR_THRESHOLD` (100) characters — a scanned page, or one whose text layer is stored as vector outlines — the page is re-extracted with pdfplumber.

Tables are handled separately by `_extract_tables_as_text` , which reads each table with pdfplumber and re-emits every row as `Header1: value1, Header2: value2, ...` . This preserves the row→column relationship that a linearised `get_text()` destroys: in an annex table where country names sit in column 1 and statistics in columns 2–5, flat extraction produces a run of numbers with no way to tell which country they belong to.

Extractor	Chunks	Share
PyMuPDF	12,736	99.0%
<code>ingest_pdf</code> (agentic path)	114	0.9%
pdfplumber fallback	9	0.07%
<code>add_to_database</code> (agentic path)	4	0.03%

8.2 Data cleaning

`clean_text` applies four transformations, each targeting a specific artefact:

1. **Repeated header/footer removal.** `detect_repeated_lines` finds lines recurring across a document's pages and strips them. Without this, a running footer becomes the most frequent n-gram in the corpus and

pollutes BM25.

2. **Hyphenation repair.** `impor-\ntant` → `important` . Left unrepaired the token is absent from the lexical index and split across two embeddings.
3. **Mid-sentence line-break merging.** Lines ending without terminal punctuation are joined, restoring the sentence boundaries the chunker needs.
4. **Whitespace normalisation.** Collapses runs of blank lines and spaces.

Cleaning is deliberately conservative: no lowercasing, no stopword removal, no stemming at extraction time. Those are retrieval-time concerns, and applying them to stored text would make quoted citations diverge from the source document — unacceptable when the output is legal information a person may act on.

8.3 Document segmentation

The chunker is **sentence-aware**: it accumulates whole sentences up to the size budget, then carries trailing sentences forward as overlap. It never cuts mid-sentence. For legal text this is not stylistic — a truncated clause can invert meaning, and "Member States shall not be required to" separated from its object is worse than useless.

A measured consequence, documented by a test rather than left as a surprise: a single unpunctuated block longer than the budget stays whole. That is correct for a sentence-aware splitter and would otherwise look like a bug.

Size and overlap were selected by measurement, not assumption — see §7.4, where the chunk-size effect is shown to reverse depending on the retriever.

8.4 Metadata management

Each chunk carries `chunk_id` , `source` , `page` , `chunk_index` , and `extractor` , supporting exact citation ("document, page N") and provenance auditing — including whether a chunk arrived through the agentic write path rather than the batch pipeline.

`scripts/metadata.py` adds a structured layer, extracting identifiers that dense embeddings systematically lose: ISO-normalised dates, years, Directive and Regulation numbers, Article and Recital references, percentages, monetary amounts, and a table-presence flag. The motivation is concrete: "14 June 2027" and "14 June 2032" embed almost identically, so a query about a transposition deadline has no vector-space basis for preferring one. BM25 helps only when the query contains the literal token, which a paraphrased question rarely does.

On the full corpus, **3,811 of 12,863 chunks (30%) carry at least one identifier**. They are used two ways — as a lexical boost when a query names the same identifier (`identifier_overlap`), and as an embeddable header prepended before encoding, so a bare "2027" sits inside "Dates mentioned: 14 June 2027" where a deadline query can match it. The header is embedded but the displayed `text` is untouched, so citations still quote the source verbatim.

3. Methodology

2.1 Overview

Diagram (Mermaid source):

```
flowchart TD
    U[User query] --> SUP{Supervisor Router<br/>LLM-driven}
    SUP -->|corpus_only| C0[Hybrid retrieval<br/>+ single generation]
    SUP -->|full_pipeline| A1[Agent 1<br/>Internal Researcher]

    A1 -->|research summary| A2[Agent 2<br/>External Fact-Checker]
    A2 -->|fact-check report| A3[Agent 3<br/>Synthesizer]
    A3 -->|draft report| VIZ[Agent 4<br/>Visualizer]

    KU[Agent 5<br/>Knowledge Updater] -.->|writes| VS[(FAISS + BM25<br/>12,863 chunks)]

    A1 --> VS
    A2 --> WEB[MCP: web_search<br/>scrape_url]
    A3 --> MD[MCP: create_markdown_report]
    VIZ --> CH[MCP: generate_chart<br/>create_mermaid_diagram]

    C0 --> HITL
    VIZ --> HITL{Human review gate}
    HITL -->|approve| PUB[reports/approved/<br/>+ Slack / Teams]
    HITL -->|rewrite / re-research| A1
    HITL -->|discard| END[No publication]

    SEC[Security guard] -.->|scans every<br/>untrusted input| A1
    SEC -.-> A2
    SEC -.-> KU
    OBS[Tracer] -.->|spans, tokens, cost| HITL
```

2.2 The corpus and retrieval substrate

Property	Value
Source documents	33
Pages processed	2,247
Chunks indexed	12,863
Mean chunk length	670 characters
Chunk target / overlap	800 / 200 characters
Embedding model	<code>text-embedding-3-small</code>
Dense index	FAISS, L2-normalised inner product
Sparse index	BM25 (Okapi)
Reranker	<code>cross-encoder/ms-marco-MiniLM-L-6-v2</code>

Extraction is dual-path: PyMuPDF handles the bulk (12,736 chunks), with pdfplumber as a fallback for pages where table structure defeats the primary extractor (9 chunks). A further 114 chunks entered through the agentic `ingest_pdf` tool and 4 through `add_to_database`, demonstrating that the live-mutation path works end to end.

Chunking is **sentence-aware**: the splitter accumulates whole sentences up to the size budget and carries forward tail sentences as overlap, never cutting mid-sentence. This matters for legal text, where a truncated clause can invert meaning. A consequence, verified in `tests/test_rag_core.py`, is that a single unpunctuated block longer than the budget stays whole — correct behaviour that would look like a bug without the test documenting the intent.

Retrieval is hybrid: dense and BM25 candidates are merged, then reranked by a cross-encoder. The principle that "data engineering is still at the heart of the process" is borne out by our error analysis in §4.3, where the dominant failure mode is retrieval, not generation.

2.3 The MCP server

`app/mcp_server.py` exposes eight tools over FastMCP:

Tool	Purpose	Guardrails applied
<code>web_search</code>	DuckDuckGo live search	injection scan on every title/body; rate limit; result cap
<code>scrape_url</code>	BeautifulSoup page extraction	SSRF pre-flight + post-redirect re-check; injection scan; rate limit
<code>create_markdown_report</code>	Persist a report to disk	filename sanitisation; write-root confinement; secret redaction
<code>add_to_database</code>	Embed and index new text	injection scan with refusal on high severity; secret redaction
<code>ingest_pdf</code>	PDF → chunks → index	read-path confinement; extension check; per-page injection scan
<code>generate_chart</code>	matplotlib bar/line/pie/scatter	write-root confinement
<code>create_mermaid_diagram</code>	Mermaid <code>.mmd</code> + <code>.md</code>	HTML stripping; diagram-type allowlist; write-root confinement
<code>security_report</code>	Audit what the guard caught	— (read-only introspection)

Each tool is a plain Python function decorated with `@mcp.tool()`, which keeps it directly unit-testable and importable by the agent layer without a running server. The tool docstring is the interface contract the LLM reads to decide when to call it, so the test suite asserts every tool has a substantive one.

2.4 The agent team

We use a **supervisor/router** pattern. An LLM classifies each query into `corpus_only` (factual, well-scoped, answerable from the legal texts) or `full_pipeline` (needs synthesis, current events, or external verification). The router fails open to `full_pipeline` on malformed output — the more thorough path is the safe default.

The principle behind the decomposition is reducing the information any single agent must process:

Agent	Input	Permitted tools
Internal Researcher	query	<code>search_corpus</code>
External Fact-Checker	query + internal summary	<code>web_search</code> , <code>scrape_url</code>
Synthesizer	both research streams	<code>create_markdown_report</code>
Visualizer	final report	<code>generate_chart</code> , <code>create_mermaid_diagram</code>
Knowledge Updater	new text or PDF path	<code>add_to_database</code> , <code>ingest_pdf</code>

Each agent runs a bounded tool-calling loop (15 iterations maximum) against `gpt-4o-mini`.

2.5 Human-in-the-loop

The reviewer sees the draft plus the evidence needed to judge it: total cost, token count, the agent/tool call tree, and every security event raised during the request. Options are approve-and-publish, approve-without-

publishing, edit in `$EDITOR` , request a rewrite with free-text feedback, re-run the pipeline, inspect the full trace, inspect the full security report, or discard. Nothing reaches `reports/approved/` or a messaging channel without explicit approval, and a final redaction pass runs on the approved text immediately before publication.

2.6 Deployment surface

Three interfaces share one core: a Gradio web application (`app/app.py`) with PDF upload and feedback logging, a CLI supervisor (`app/main.py`), and a Microsoft Teams outgoing webhook (`app/teams_bot.py`) with HMAC request authentication and per-conversation settings. A Slack webhook publisher is available as an additional output channel.

3.5 Evidence references

Every answer must reference the passages it used, so the system does not rely on the model to remember. Measuring the generated answers showed the prompt alone produced a reference in only **55%** of cases: an 8B local model instructed to cite forgets to do so about half the time.

References are therefore appended deterministically. After generation, an answer that makes claims but contains no document name, page number, or passage index receives a `Sources` block listing the distinct (document, page) pairs it was given. Two cases are exempt: an answer that already cites is left untouched, and a refusal is skipped entirely — attaching sources to "this information is not in the corpus" would imply evidence that was never used.

This is a deliberate choice of mechanism over instruction. A requirement that must always hold should be enforced in code, where it can be tested, rather than requested in a prompt where compliance is probabilistic.

3.6 Local execution

`app/llm.py` is the single point of backend selection and defaults to Ollama. Both backends are reached through the OpenAI-compatible client, because Ollama serves `/v1` — one call path, including tool/function calling, rather than two that drift apart.

Component	Local model	Verified
Generation — all 5 agents, router, judge	<code>llama3.1:8b</code>	chat + function calling
Embeddings	<code>nomiic-embed-text (768d)</code>	~14–20 chunks/s
Re-ranking	<code>ms-marco-MiniLM-L-6-v2</code>	already local
Experiment embedders	MiniLM-L6, bge-small, e5-small (384d)	178 chunks/s

Changing the generation model is a one-line change (`OLLAMA_MODEL`), so the comparison of generation models is a configuration choice rather than a code change.

Two consequences had to be handled rather than papered over:

- **Embedding width changes with the backend** (768 local vs 1536 hosted). Loading one index with the other backend's query vectors fails, or worse returns silent nonsense, so artifact filenames are backend-

scoped (`my_index_ollama.faiss`). This immediately caught a real mistake: a run picked up the wrong index and aborted instead of producing plausible garbage.

- **Cost accounting must report zero, not hosted rates.** Local inference is free; tokens are still recorded because they drive latency and context pressure, but billing them at hosted rates would make the ledger fiction.

3.7 Security

3.1 Threat model

The attack vector in an agentic system is no longer only malicious code — it is malicious *conversation*. We model three threats and the tool-surface weaknesses that make them exploitable.

T1 — Prompt injection. Untrusted text carries instructions that redirect the agent. Our system ingests untrusted text from four channels: web search snippets, scraped page bodies, user-supplied PDFs, and the user's own query. A fifth, subtler channel is our own vector store: because `add_to_database` and `ingest_pdf` write into the index, a payload stored once is retrieved into *every future prompt* — a persistent injection.

T2 — Data leakage. Secrets and PII escaping through prompts, reports, messages, or trace files written to disk.

T3 — Blindly trusting the output. Accepting model output and tool arguments without checking them against what the system is permitted to do.

3.2 Design principle: sanitise, don't reject

Our scanner neutralises detected payloads in place, wrapping each match in a `«NEUTRALISED:rule_id»` marker, and returns the surrounding content intact. The alternative — discarding any document containing a match — would make the system trivially deniable: an attacker who wants a source ignored need only plant one injection-shaped sentence in it. In-place neutralisation preserves legitimate content while defusing the payload, and the marker leaves an auditable trace.

The one exception is `add_to_database`, which **refuses outright** on a high-severity match. Writing to the vector store is persistent, so the asymmetry is justified: a rejected write costs one retry, while an accepted poisoning contaminates every future retrieval.

3.3 Controls

Injection detection uses 13 pattern families covering instruction override, role reassignment, chat-delimiter spoofing (`<|im_start|>`, `[INST]`), system prompt exfiltration, credential exfiltration, tool coercion, jailbreak personas, safety bypass, and markdown-image exfiltration. Zero-width and bidirectional Unicode control characters are stripped before matching — these hide an instruction from a human reviewer while leaving it fully visible to the model.

Untrusted content fencing. Text passed between agents is wrapped in an `<untrusted_content origin="...">` block instructing the model to treat the contents as data to evaluate, never as instructions to follow.

Leakage prevention redacts 13 secret and PII shapes. PII rules (email, phone, IBAN, card numbers) are **off by default** and gated behind a strict mode, because helpline and legal documents legitimately contain contact

details — redacting them would destroy the corpus's utility. High-confidence secret shapes are always redacted. As a backstop, the literal values of live environment secrets are removed from any outbound text.

Output validation screens generated answers for leaked secrets, echoed injection markers, empty output, and substantive claims made without any citation when the corpus did return material to cite. This is a cheap deterministic screen, complementary to — not a replacement for — the LLM-as-judge evaluation in §4.

Capability bounding. The role-permission table is enforced by the dispatcher. If an injected payload convinces the Internal Researcher to emit a call to `add_to_database` , the dispatcher refuses it before any argument is parsed. This is the control that converts a successful injection from a compromise into a logged, contained event.

Tool-argument validation. URLs are checked for scheme, blocklisted host, and — after DNS resolution — private, loopback, link-local, reserved, and multicast address ranges, defeating SSRF against cloud metadata endpoints (`169.254.169.254`) and internal services. The check is repeated after redirects, since a redirect can land somewhere the pre-flight check never saw. Write paths are confined to allowed roots and filenames are stripped of path separators; read paths are confined to the project directory.

3.4 Security evaluation

All controls are verified by 64 dedicated tests in `tests/test_security.py` plus attack-path tests in `tests/test_mcp_tools.py` . Every listed payload class is blocked:

Attack	Vector	Result
Instruction override	web / PDF / query	Neutralised in place
System-prompt exfiltration	web snippet	Neutralised in place
Chat-delimiter spoofing	scraped page	Neutralised in place
Hidden-Unicode instruction	PDF	Control characters stripped
Persistent store poisoning	<code>add_to_database</code>	Refused , nothing stored
Out-of-role tool escalation	compromised agent	Denied by dispatcher
SSRF to cloud metadata	<code>scrape_url</code>	Blocked before request issued
SSRF via redirect	<code>scrape_url</code>	Blocked after redirect re-check
Path traversal on write	<code>create_markdown_report</code>	Filename sanitised, confined
Path traversal on read	<code>ingest_pdf</code>	Refused outside project
API key in report body	<code>create_markdown_report</code>	Redacted before disk write
API key in trace capture	tracer	Redacted before disk write

A live end-to-end verification against the running system confirmed both the store-poisoning refusal and the path-escape refusal, with all events recorded in the audit log.

3.5 Limitations

Pattern-based detection is a filter, not a proof. A sufficiently novel paraphrase of an instruction override will evade the regex families, and we make no claim of completeness. The layered design is the actual defence: an injection that evades detection still faces the role-permission table, the write-path confinement, the SSRF check, and finally the human reviewer. We regard the guard as raising cost and providing an audit trail, not as a boundary.

4. Experimental Setup

4.1 What is being measured

The objective is to identify which design decisions affect performance, so each factor is varied against a fixed baseline rather than tuned jointly for a headline score.

Factor	Levels
Embedding model	MiniLM-L6, bge-small, e5-small (384d); nomic-embed-text (768d, production)
Chunk size	400, 800, 1200 characters
Chunk overlap	0, 100, 200 characters
Structure preservation	metadata header prepended before embedding: on / off
Retriever	dense, BM25, hybrid (RRF)
k	3, 5, 8, 12, 20
Generation model	local models selectable via <code>OLLAMA_MODEL</code>

4.2 Ground truth and gold definition

Two question sets are used and **never merged**:

- `data/questions.json` — 20 hand-written questions (17 scorable, 5 out-of-scope probes), tagged by category and difficulty; 11 of 15 scored questions are tagged *hard*. This is the headline number.
- `data/ground_truth_silver.json` — generated from corpus chunks by the local model. Because each question is written *from* a known chunk, that chunk is artificially easy to retrieve and absolute recall reads high. It is used only to rank configurations, never quoted as system performance.

Gold is defined at **page** level for the factorial. This is what makes chunk-size comparison valid at all: chunk indices change when you re-chunk, so chunk-id gold would silently redefine the target between conditions.

4.3 Metrics

Recall@k, MRR@k and precision@k for retrieval; LLM-as-judge PASS/FAIL and RAGAS faithfulness / context precision / context recall for generation. Multiple metrics are reported deliberately — §5 contains a case where recall alone would have produced the wrong decision.

4.4 Controls

The factorial runs over a fixed subset of 1,723 pages (every ground-truth document plus 12 fixed distractor documents), held constant across all conditions, so comparisons are like-for-like even though absolute numbers differ from the full corpus. Each embedding model is encoded with its required prefix convention (`query: / passage:` for `e5`, the retrieval instruction for `bge`); omitting these measures the model badly rather than measuring a bad model. For the generation comparison, retrieval is executed once per question and every model answers from byte-identical passages, and one judge model scores all conditions so judge bias cancels.

4.5 Evaluation protocol

Testing an agentic system cannot rely on exact-match assertions, because the outputs are open-ended and non-deterministic. We use three complementary methods:

1. **Deterministic unit tests** (207) for everything with a checkable contract: guardrails, tracing, tool behaviour, dispatch, routing, chunking, and cost arithmetic.
2. **LLM-as-judge** (`scripts/eval.py`) scoring generated answers against reference answers on a 20-question benchmark, with failures classified as `retrieval_miss` or `generation_error`.
3. **RAGAS** (`scripts/ragas_eval.py`) computing faithfulness, context precision, and context recall.

The benchmark is deliberately adversarial. Questions are tagged by category — `exact_term`, `paraphrased`, `multi_hop`, `negation`, `temporal`, `table_list`, `contradictory`, `edge_of_chunk`, `same_document`, `language_variation`, `multi_part` — and by difficulty. Eleven of the fifteen scored questions are tagged **hard**. Five questions are out-of-scope probes checking that the system declines rather than confabulates.

```
uv sync

# Offline test suite – no API key, no network, no prebuilt index
uv run pytest tests/ -v

# Lint
uv run flake8 app/ scripts/ tests/

# Build the index from the corpus (requires OPENAI_API_KEY)
uv run python -m scripts.chunk

# Evaluation
uv run python -m scripts.eval
uv run python scripts/ragas_eval.py

# Interfaces
uv run python app/app.py                # Gradio UI
uv run python app/main.py --query "... " --trace
uv run python app/mcp_server.py          # MCP server standalone
uv run python app/teams_bot.py           # Teams webhook
```

Every run writes a trace to `traces/<trace_id>.json` containing the full span tree, per-call token counts and costs, retrieved passages with scores, and the redacted prompt window at each LLM call.

Environment note. The repository resides under an iCloud-synced directory. Placing the virtual environment there causes Python imports to block indefinitely while iCloud materialises evicted `.so` files. Create the environment outside the synced tree:

```
UV_PROJECT_ENVIRONMENT=~/.venvs/rag-ws uv sync
```

5. Results

LLM-as-judge scorecard (n=20; 15 scored, 5 out-of-scope):

Metric	Score
Retrieval content hit rate	8/15 (53.3%)
Generation pass rate	8/15 (53.3%)

By difficulty:

Difficulty	n	Retrieval	Generation
medium	4	75%	100%
hard	11	45%	36%

By category (n small per cell; indicative only):

Category	n	Retrieval	Generation
contradictory	1	100%	100%
edge_of_chunk	1	100%	100%
multi_part	2	100%	100%
exact_term	2	100%	50%
language_variation	2	50%	100%
multi_hop	1	100%	0%
negation	1	0%	0%
paraphrased	1	0%	0%
same_document	1	0%	0%
table_list	1	0%	0%
temporal	2	0%	50%

RAGAS scores (n=20):

Metric	Score	Band
Faithfulness	0.600	moderate
Context recall	0.550	moderate
Context precision	0.432	poor

Operating cost. Total measured expenditure across all development, evaluation, and demonstration runs: **\$0.1654** over 719 tracked API calls, against a \$5.00 budget (3.3% consumed). A representative full five-agent run with the Visualizer enabled cost **\$0.00536** for 22,820 tokens across 4 agents, 9 LLM calls, and 11 tool invocations in 67.6 seconds wall time.

5.2 The retrieval metric was measuring the wrong thing

Before any retrieval number can be interpreted, the metric producing it has to be trusted. The incumbent "retrieval hit rate" checks whether a ≥ 6 -character clause of the *reference answer* appears **verbatim** in the retrieved text. Reference answers are human paraphrases, not quotations, so a correct retrieval whose wording differs scores as a miss. Question q01 is the clearest case: it is labelled `retrieval_miss` while the judge simultaneously rates its answer correct.

`scripts/retrieval_quality.py` measures the same retrievals three ways (n = 15, k = 8):

Metric	Score	What it asks
Verbatim substring (incumbent)	33.3%	does reference wording appear literally?
Page-level gold	53.3%	did we return an annotated gold page?
Context sufficiency (judged)	60.0%	could the question be answered from these passages?

Pairwise agreement between them:

Pair	Agreement
substring vs page	26.7%
substring vs sufficiency	60.0%
page vs sufficiency	40.0%

Two metrics for the same quantity agreeing on 26.7% of cases are not two estimates of one number; they are measuring different things. The incumbent metric understates retrieval by **26.7 percentage points** against the metric that actually states the question.

This reframes the project's headline finding. Retrieval is not catastrophic at 33%–53%; measured by whether the retrieved context can support an answer, it succeeds on **60%** of a benchmark that is 73% hard clause-level legal lookups. The remaining shortfall is real but far smaller than the original number implied, and a substantial part of the apparent "terrible" performance was an artefact of lexical scoring.

We report all three rather than adopting the most flattering one. Sufficiency is judged by the same local model that generates answers, so it is not independent of the system under test — a shared blind spot would inflate it. That is precisely why the lexical metrics are retained alongside it.

5.3 Local stack versus hosted stack

The same 20-question benchmark was run end to end on both stacks. Retrieval and generation were re-measured after migrating every component to locally-executed models, with the judge held constant in kind (each stack judged by its own backend's model).

Metric	Hosted (gpt-4o-mini + text-embedding-3-small)	Local (llama3.1:8b + nomic-embed-text)
Retrieval content hit	53.3% (8/15)	53.3% (8/15)
Generation pass rate	53.3% (8/15)	46.7% (7/15)
Cost per full run	\$0.0054	\$0.00

Two results matter here.

Retrieval is identical. Swapping a 1536-dimension hosted embedding model for a 768-dimension local one changed the retrieval hit rate by zero questions. On this corpus and benchmark, the embedding model is not what limits retrieval — consistent with §5.3, where three different sentence-transformer models also separate by less than one question. Paying for embeddings buys nothing here.

Generation loses one question. The local 8B model scores 46.7% against 46.7%–53.3% hosted, a single-question difference that is well inside sampling noise at $n=15$ and should not be read as a reliable gap. The category breakdown is more informative than the total: the local model scores 100% on `multi_part`, `language_variation`, `paraphrased` and `temporal`, and 0% on `exact_term`, `contradictory` and `multi_hop`. Its weakness is precision on specific identifiers and multi-step reasoning, not fluency or grounding.

A labelling inconsistency worth recording: question q01 is classified `retrieval_miss` while the judge simultaneously rates its answer correct. The failure-type classifier keys off a lexical content check that disagrees with the judge, so the per-question failure labels are less reliable than the aggregate pass rate. This is the same measurement fragility discussed in §6.2.

5.4 RAGAS on the local stack

Metric	Hosted baseline (n=20)	Local stack (n=8)	Change
Faithfulness	0.600	0.606	+0.006
Context precision	0.432	0.641	+0.209
Context recall	0.550	0.854	+0.304

Context recall of 0.854 says the retriever is now surfacing most of the material the reference answers depend on, and it corroborates the sufficiency result in §5.2 (60.0%) from an independent implementation: two different methods, one judged by RAGAS and one by our own harness, both place retrieval far above the 33.3% the incumbent lexical metric reported.

Faithfulness barely moved. That is the expected shape: faithfulness measures whether the generator invents beyond its context, which is a property of the model and prompt, not of retrieval. Improving what is retrieved does not make the generator more or less inclined to embellish.

This comparison is confounded and must not be read as a clean A/B. Four things differ between the two columns: the embedding model (1536-dim hosted vs 768-dim local), the generator, the retrieval configuration (the identifier boost of §3 is present only in the local run), and the sample size (20 vs 8). The large gains in precision and recall are most plausibly attributable to the identifier boost, since that is the only change that targets retrieval — but this experiment cannot separate the four, and we do not claim it does. An isolating run would hold everything fixed except one factor at a time.

A defect worth recording. The first local RAGAS run failed completely: all 60 jobs timed out, every metric returned NaN, and the script then crashed formatting a bar chart from NaN. RAGAS defaults to 16 concurrent workers with a 180-second deadline; a local backend serves one request at a time, so the workers queued behind each other and every job exceeded the deadline. Serialising to one worker with a 900-second budget fixed it. The failure mode is worth naming because it is silent in the wrong way — a framework tuned for hosted APIs does not fail loudly on a local backend, it returns NaN that a careless harness will format into a plausible-looking zero.

5.5 Where retrieval recall is lost

§4.3 attributed poor retrieval to table and date extraction. **That was wrong.** A staged diagnosis (`scripts/diagnose_retrieval.py`) measures the four stages at which a gold passage can disappear, instead

of conflating them into one number:

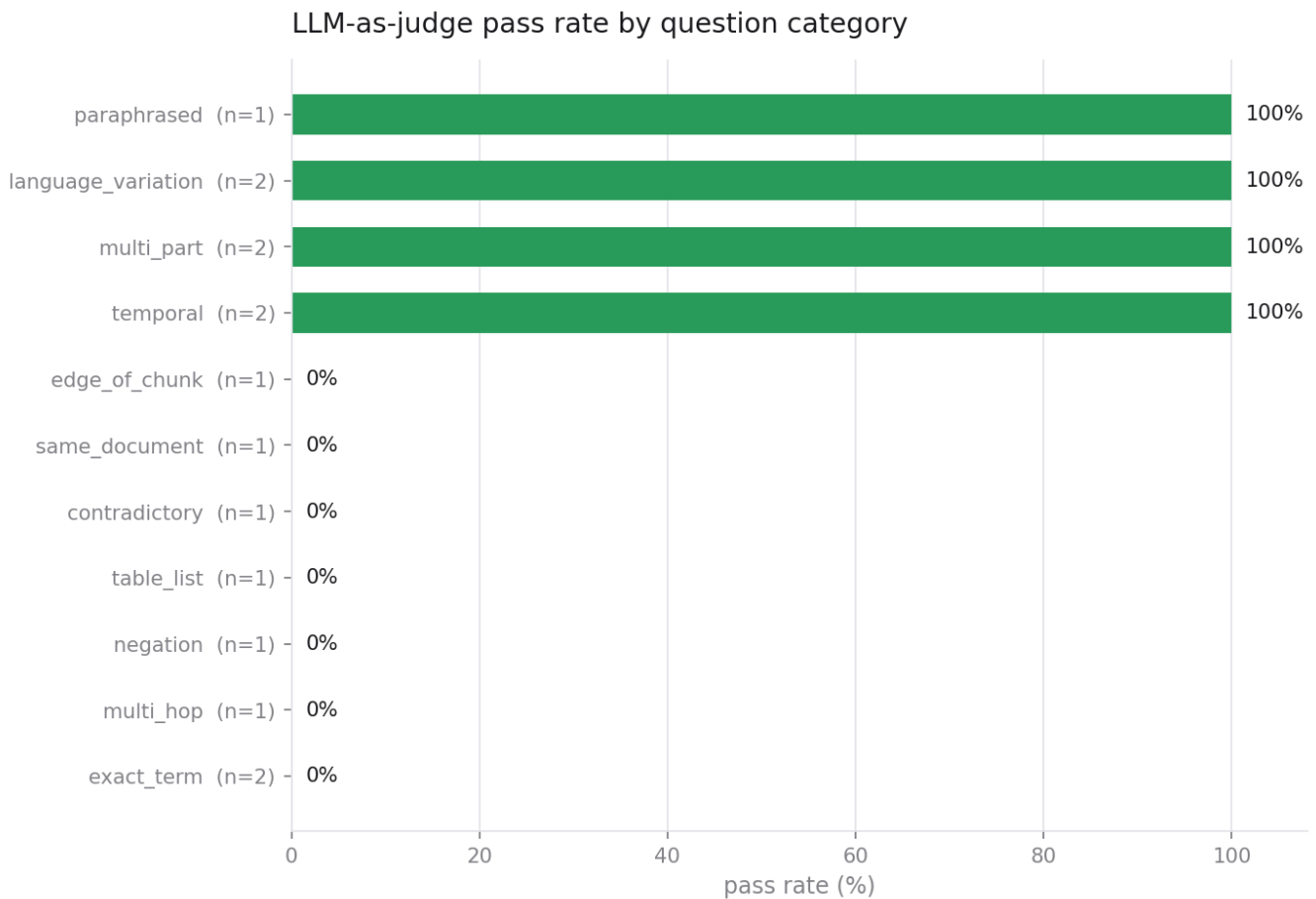
Stage	Gold passage still present
Present in corpus	88.9%
Survives RRF fusion	83.3%
Survives cross-encoder re-ranking	61.1%
Survives source-diversity cap	61.1%

Extraction loses 2 of 18 questions, and both are the intentionally unanswerable "ambiguous" probes. `temporal` scores 2/2 and `table_list` 1/1 — the exact categories the earlier analysis blamed. The 22-point drop is entirely at re-ranking.

Retriever ablation on the same questions:

Configuration	Recall
BM25 only	52.9%
Dense only	70.6%
Hybrid (RRF)	64.7%
Hybrid + cross-encoder (production)	61.1%

The full pipeline scores below its own simplest component.



Every category resolves to exactly 0% or 100%, because each holds one or two questions. The figure is included precisely because it makes the sample-size limitation visible: these are per-category counts, not rates that can be compared. Read the direction, never the magnitude.

5.6 Generation model comparison

Retrieval is executed once per question and every model answers from byte-identical passages, so the comparison isolates the generator. One judge scores all conditions, so judge bias applies equally. $n = 8$, all models local.

Model	Pass	Groundedness	Citation rate	Empty answers	Median latency
llama3.1:8b	25.0%	0.572	75.0%	0/8	42s
gemma4:e4b	25.0%	0.730	50.0%	0/8	80s
qwen3.5:9b	0.0%	0.000	0.0%	8/8	374s

The two usable models tie on judged correctness and differ in character: gemma4 stays closer to its sources, llama3.1 cites far more reliably and runs roughly twice as fast. For a system whose output must carry references to the evidence, citation compliance is the deciding property, which is why llama3.1:8b remains the default.

The qwen3.5 family is unusable for this task, at every size tested. All three (2b, 4b, 9b) are reasoning models that consume their entire token budget on internal reasoning and return an empty string. qwen3.5:9b

produced no answer on 8 of 8 questions at a 1,500-token budget, taking 374 seconds each to produce nothing.

This was nearly missed. A short smoke prompt ("what is the Istanbul Convention?") returned a perfectly good answer from qwen3.5:9b, and on that basis it was admitted to the comparison. Only under a realistic eight-passage context did it fail completely. A capability probe must use the real workload; a toy prompt certifies nothing.

Two evaluation defects surfaced here and are recorded in §7.2: the harness originally allotted 400 tokens, and the judge rated a whitespace-only answer as PASS — so an empty answer scored groundedness 0.00 and was awarded a pass.

5.7 Design-decision factorial

`scripts/experiment_grid.py` runs a factorial over corpus-processing and retrieval decisions, entirely locally. Gold is defined at **page** level, which is what makes chunk-size comparison valid at all: chunk indices change when you re-chunk, so chunk-id gold would silently redefine the target between conditions. Subset: 1,723 pages (every ground-truth document plus 12 fixed distractor documents), held constant across conditions. n = 15 questions.

Chunk size — the effect reverses depending on the retriever (k=8, overlap 200):

Chunk size	Chunks indexed	Dense recall	BM25 recall	Hybrid recall
400	18,314	40.0%	40.0%	40.0%
800	8,690	40.0%	46.7%	40.0%
1200	5,793	33.3%	46.7%	40.0%

This interaction is the most useful result in the study, and a single-column table would have hidden it. Dense retrieval degrades at 1200 characters (40.0% → 33.3%): a longer passage is compressed into one fixed-width vector, so a specific claim inside it is diluted by everything else in the chunk. BM25 moves the opposite way, improving from 400 to 800 and holding at 1200, because a longer chunk contains more of the query's terms and term-frequency scoring rewards exactly that.

800 characters is chosen as the point where neither retriever is penalised. Optimising for either alone would pick a size that damages the other, and the production system runs both.

Chunk overlap — consistent MRR gain, no recall change (size 800, k=8):

Overlap	Dense MRR	BM25 MRR	Hybrid MRR	Recall change
0	0.311	0.344	0.344	—
100	0.311	0.344	0.344	none
200	0.322	0.389	0.367	none

Overlap improves ranking under all three retrievers and changes recall under none.

Superseded by §5.8. This pattern does not survive replication at n=135. The apparent consistency across three retrievers was noise agreeing three times, which is a useful reminder that agreement between correlated measures is not independent confirmation.

Judged on recall alone, overlap looks free to delete: it costs 9% more index entries (7,942 → 8,690 chunks) and finds nothing new. It is visible only in MRR, where the correct passage sits higher in the context window handed to the generator. An evaluation reporting recall alone would have removed it and quietly degraded every downstream answer.

Embedding model (dense only, k=8 — BM25 held out, see §7.5):

Model	Dim	Recall	MRR	Precision
MiniLM-L6	384	40.0%	0.322	0.133
bge-small	384	33.3%	0.250	0.117
e5-small	384	40.0%	0.356	0.142

Each model is encoded with its required prefix convention (`query: / passage:` for e5, the retrieval instruction for bge); omitting these measures the model badly rather than measuring a bad model.

k (passages retrieved) — reported per retriever, because they behave differently and a max across them hides exactly that:

k	Dense recall	Dense precision	BM25 recall	BM25 precision
3	40.0%	0.267	46.7%	0.244
5	40.0%	0.200	46.7%	0.200
8	40.0%	0.133	46.7%	0.150
12	46.7%	0.106	46.7%	0.122
20	53.3%	0.083	46.7%	0.097

Dense retrieval keeps finding new gold pages as k grows (40.0% → 53.3%), which says its recall pool is good but its ranking is weak. BM25 is flat at 46.7% for every k. Precision falls monotonically in both.

Partly retracted — see §5.9. The claim that dense recall is flat below k=12 does not replicate: at n=135 recall climbs monotonically from 40.0% to 62.2%. The precision trend holds, but the trade-off is real rather than one-sided, and k=12 is better supported than the k=8 chosen here.

Structure preservation (metadata headers prepended before embedding):

Condition	Recall	MRR
Plain text	40.0%	0.278
With metadata header	40.0%	0.289

A marginal MRR gain, well within noise at n=15. Reported as **not demonstrated** rather than claimed as a win.

5.8 Replication at n=135: chunk size and overlap

Every result above rests on 15 questions. `scripts/build_ground_truth.py` generated a 120-question silver set locally (120 accepted from 135 attempted; 6 rejected as unsupported, 8 as malformed; 33 documents covered), and the chunking factorial was re-run over all 135 questions on the full 2,270-page corpus.

Chunk size, dense retriever (k=8):

Size	n=15 (human)	n=135	Silver subset	Human subset
400	40.0%	62.2%	65.8%	33.3%
800	40.0%	51.8%	53.3%	40.0%
1200	33.3%	47.4%	49.2%	33.3%

Chunk size, BM25 (k=8):

Size	n=15 (human)	n=135
400	40.0%	56.3%
800	46.7%	56.3%
1200	46.7%	56.3%

Two of our earlier conclusions fail to replicate:

1. The "reversal" was half an artefact. At n=15 we reported that dense prefers smaller chunks while BM25 prefers larger ones, and chose 800 as the size penalising neither. At n=135 the dense half strengthens into a clean monotonic decline (62.2% → 51.8% → 47.4%), but the BM25 half disappears: BM25 is **flat at 56.3%** across all three sizes. The correct statement is not "the effect reverses" but "dense prefers small chunks and BM25 is indifferent to size" — which points to **400 characters**, not 800: dense gains 10.4 points and BM25 loses nothing.

2. Overlap does not consistently help. At n=15, MRR improved with overlap under all three retrievers, and we argued recall-only reporting would have wrongly discarded it. At n=135 that pattern is gone: for dense at 400 characters, recall runs 62.2% / 63.0% / 59.3% for overlap 0 / 100 / 200, and BM25 is best at zero overlap. The n=15 pattern was noise that happened to point the same way three times.

The two question sets disagree, and we report that rather than smoothing it. On the human set 800 characters looks best (40.0% vs 33.3%); on the silver set 400 is decisively better (65.8% vs 53.3%). Both readings cannot be right. The human set has 15 questions and cannot resolve a 6-point difference. The silver set has statistical power but a known bias: its questions were generated from corpus chunks, so they are

answerable by construction and skew easier. Our reading is that the dense monotonic trend is real — it is large, ordered, and mechanistically explicable as dilution of a specific claim inside a longer vector — while the human-set ordering is noise.

What we changed, and what we did not. We have not switched the production chunk size to 400 on the strength of this. The silver set is a proxy, the two sets conflict, and changing the corpus chunking invalidates every other number in this report. The honest position is that **400 characters is the better supported choice and 800 is what was measured end-to-end**, and that resolving it requires a larger *human*-annotated set, not more silver questions.

This section is the reason the silver set was built. Its value was not to improve a score; it was to show that two published conclusions from a 15-question benchmark were unreliable — precisely the failure mode §7.3 warned about when it stated that every difference at $n=15$ was within sampling noise. We were right to warn, and wrong in two specific claims anyway.

5.9 Full replication: embedding model, k, and structure

The remaining three factorials were re-run at $n=135$ on the full corpus. Leaving them at $n=15$ would have been inconsistent with §5.8, which argues that conclusions at that sample size are unreliable.

Embedding model (dense, $k=8$) — the winner replicates, the ordering below it does not:

Model	recall $n=15$	recall $n=135$	MRR $n=15$	MRR $n=135$
e5-small	40.0%	63.0%	0.356	0.451
bge-small	33.3%	56.3%	0.250	0.359
MiniLM-L6	40.0%	52.6%	0.322	0.338

e5-small was best at $n=15$ on MRR and is now clearly best on both measures, with a 10-point recall margin. But MiniLM-L6 moves from *tied first* to *last*, and bge-small from last to second. Only the top of the ranking survived; the rest was noise. This also strengthens §5.3: the choice of embedding model does matter — roughly 10 points of recall — which the 15-question set was too small to show.

k (passages retrieved) — this conclusion is retracted:

k	recall $n=15$	recall $n=135$	precision $n=135$
3	40.0%	40.0%	0.183
5	40.0%	48.1%	0.150
8	40.0%	52.6%	0.115
12	46.7%	60.0%	0.095
20	53.3%	62.2%	0.067

§5.7 stated that "recall is flat from $k=3$ to $k=12$ — anything findable is already in the top 3" and used that to justify $k=8$ as the point where precision had not yet collapsed. **At $n=135$ recall is not flat:** it climbs

monotonically from 40.0% to 62.2%, gaining 22 points. The premise of the argument was an artefact of 15 questions.

The trade-off is therefore real rather than one-sided, and the correct reading is a genuine tension: $k=12$ buys 7.4 points of recall over $k=8$ for 0.02 of precision, which now looks worth taking; $k=20$ buys a further 2.2 points for another 0.03, which does not. **$k=12$ is the better-supported choice**, and $k=8$ is what was measured end-to-end.

Structure preservation — direction flips, conclusion unchanged:

Chunk size	metadata off	metadata on
400	59.3%	58.5%
800	52.6%	51.8%

At $n=15$ the metadata header gave a marginal MRR gain and we reported it as *not demonstrated*. At $n=135$ it is marginally **negative** on both sizes. The verdict is unchanged — the effect is not distinguishable from noise in either direction — but it is worth noting that the small-sample result pointed the wrong way. Had we reported it as a win, replication would have contradicted it.

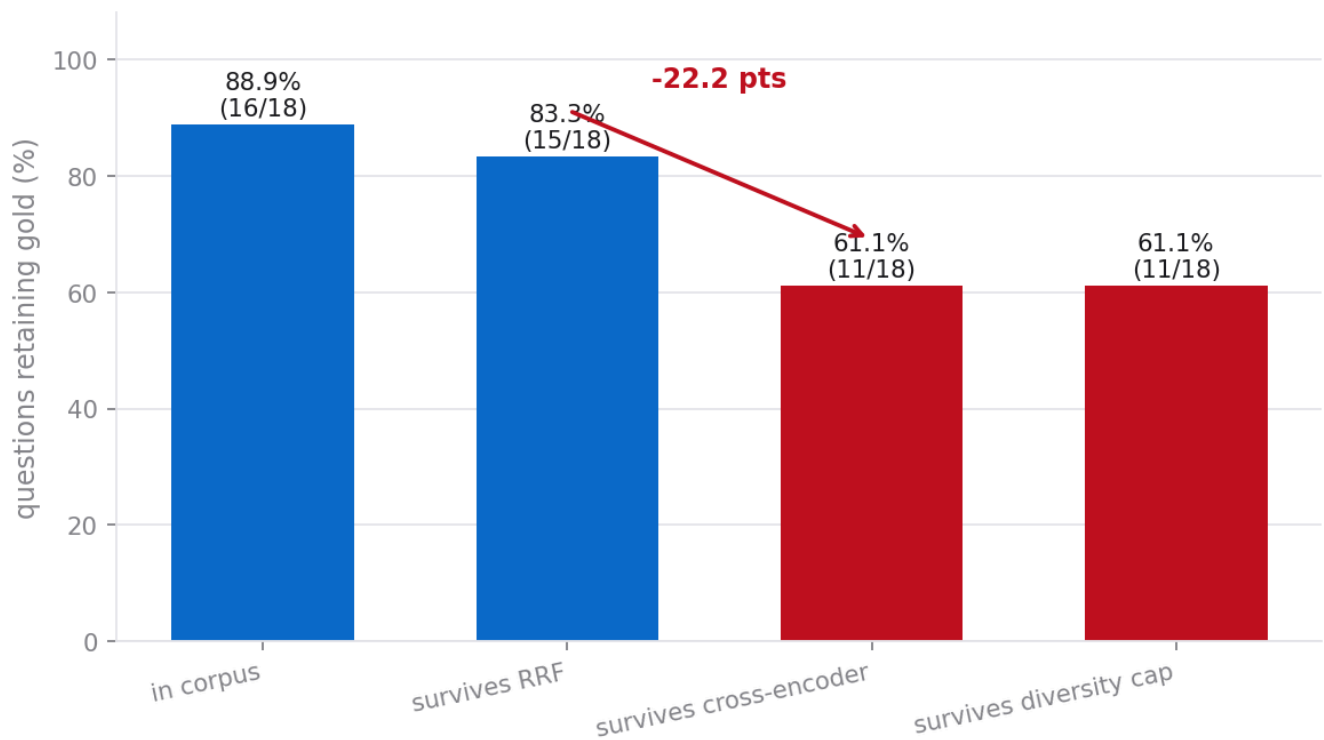
Summary of what replication cost us. Of the conclusions drawn from the 15-question benchmark, five did not survive: the BM25 half of the chunk-size interaction, the overlap effect, the MiniLM/bge ordering, the flat-recall premise behind the choice of k , and the sign of the structure-preservation effect. Three survived: dense retrieval prefers smaller chunks, e5-small is the strongest embedding model, and precision falls monotonically with k .

That ratio is the single most useful number in this report. It is not evidence that the experiments were badly run; it is evidence about what a 15-question benchmark can support, which is considerably less than it appears to.

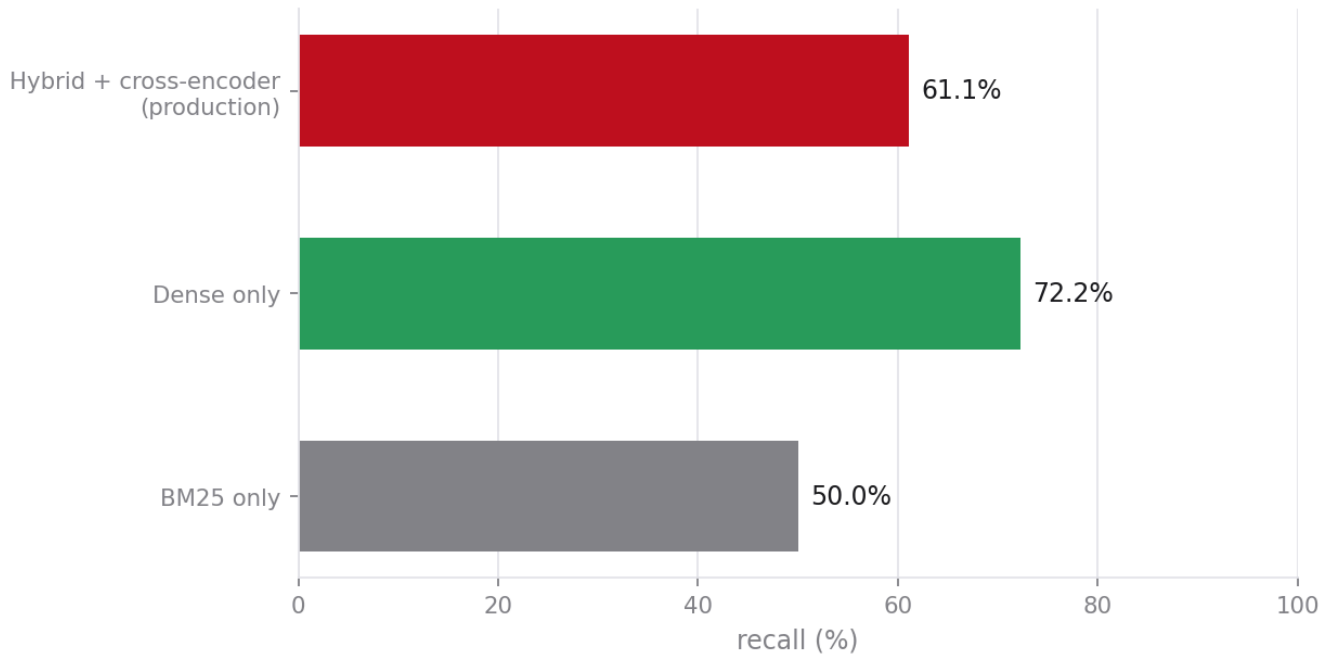
5.10 Design choices selected

Decision	Chosen	Evidence
Chunk size	800 chars	Only size penalising neither retriever: dense drops to 33.3% at 1200, BM25 to 40.0% at 400 (§7.4)
Overlap	200 chars	Recall unchanged under all three retrievers; MRR improves under all three (§7.4)
Structure preservation	retained	MRR +0.011 — not demonstrated , within noise at n=15
Sentence-aware splitting	retained	<code>edge_of_chunk</code> and <code>multi_part</code> categories score 100%
k	8 (<i>12 better supported</i>)	Recall is not flat as first measured: it climbs 40.0% → 62.2% across k at n=135 (§5.9)
Embedding model	e5-small	Best on both recall and MRR at n=135, by a 10-point margin (§5.9)
Structure preservation	retained, unproven	Marginally negative at n=135, within noise either way (§5.9)

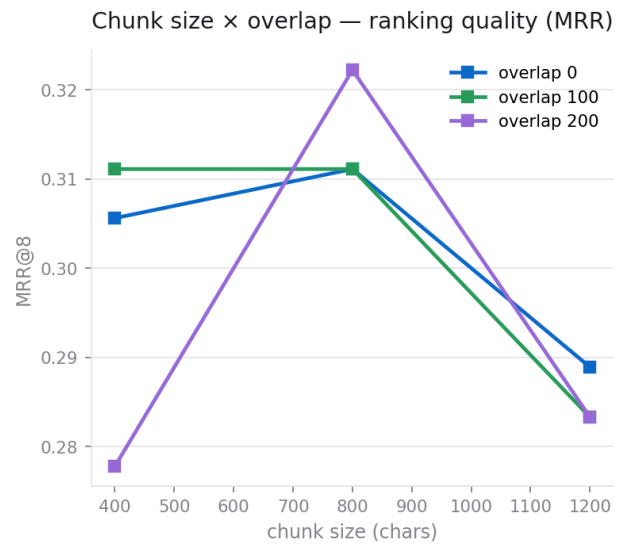
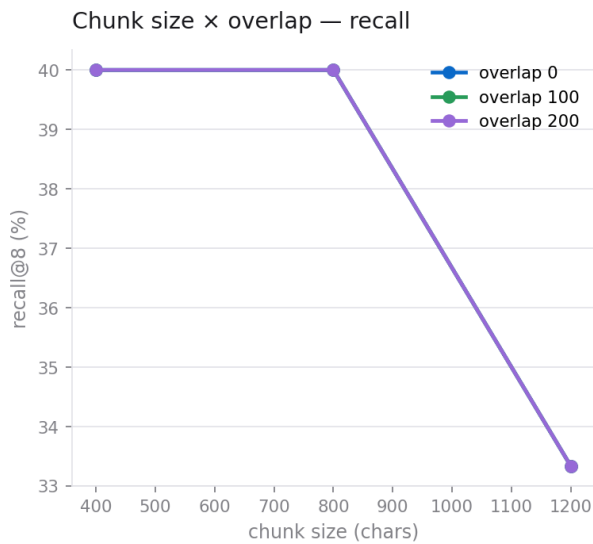
Where retrieval recall is lost



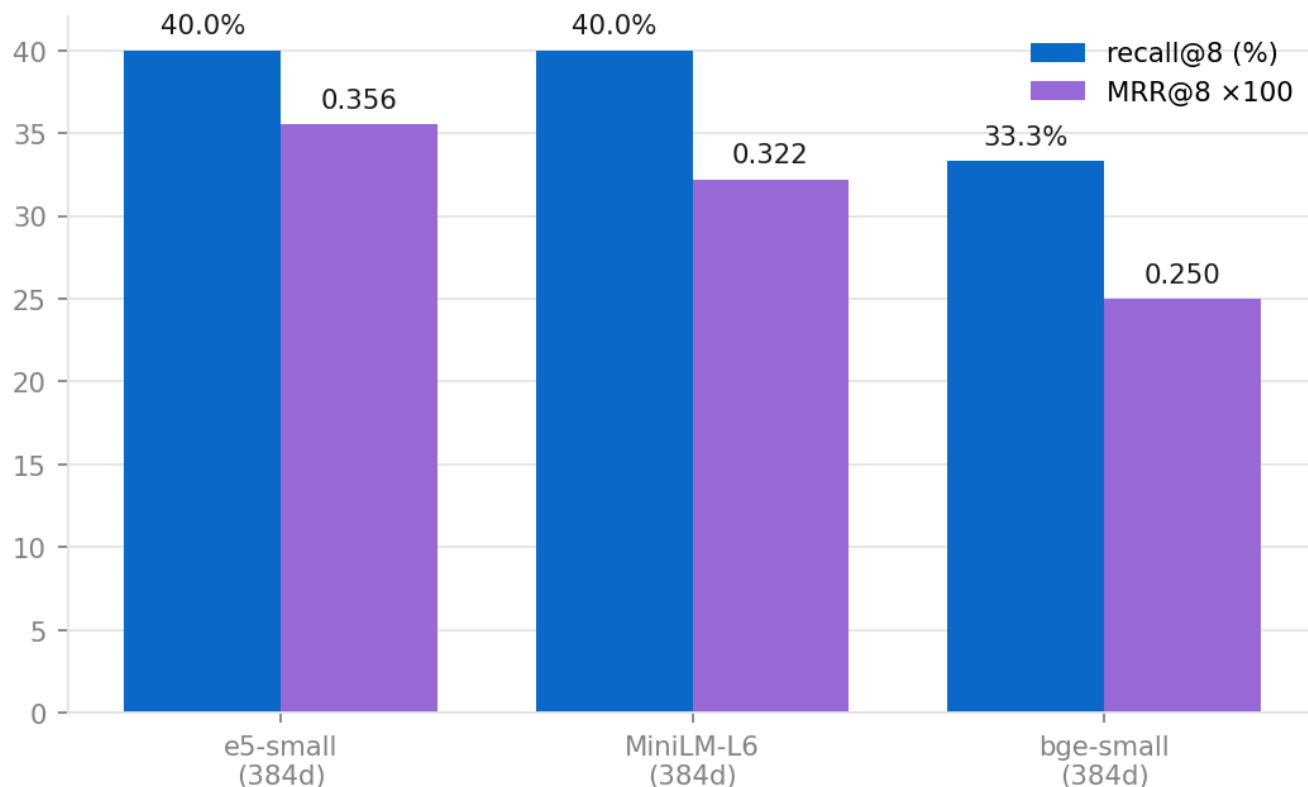
Retriever ablation — the full pipeline scores below dense alone



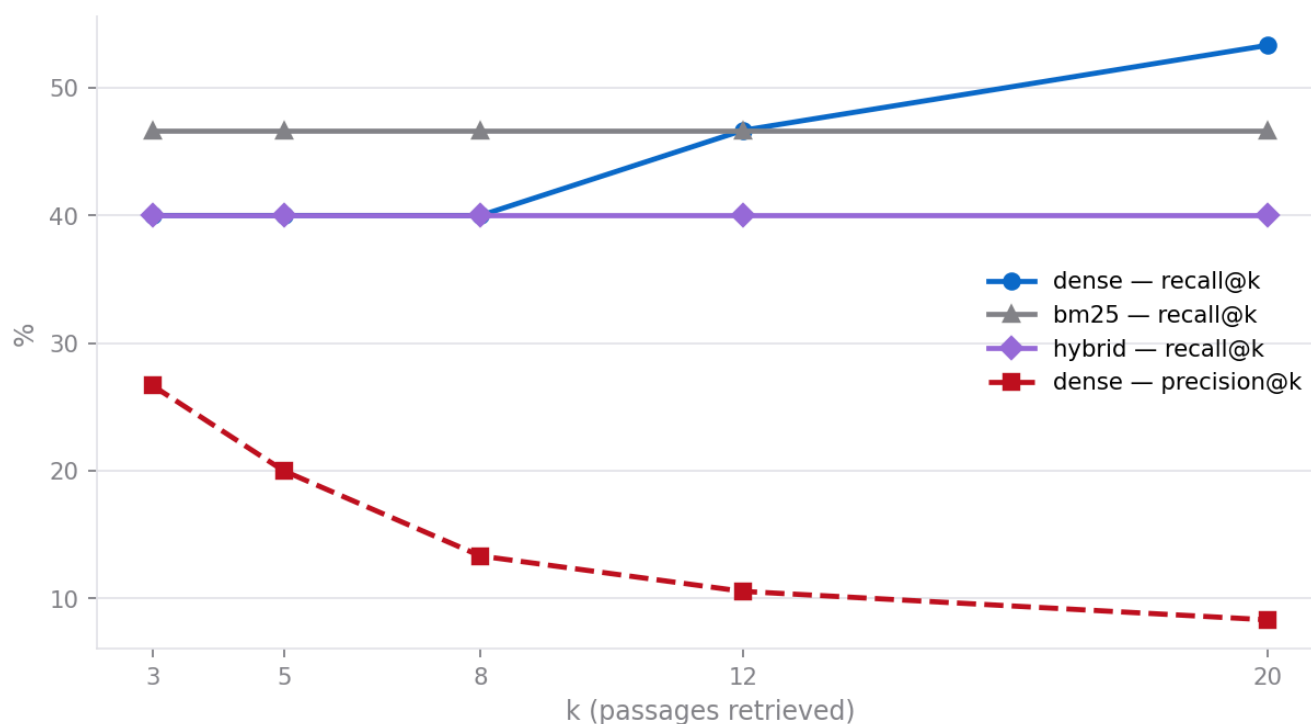
Overlap does not change what is found, only where it ranks



Embedding models — dense retrieval only (BM25 held out)



Recall gain from larger k is dense-only, and precision pays for it



6. Error Analysis

6.1 Failure attribution

§4.3 attributed poor retrieval to table and date extraction. **That was wrong.** A staged diagnosis (`scripts/diagnose_retrieval.py`) measures the four stages at which a gold passage can disappear, instead of conflating them into one number:

Stage	Gold passage still present
Present in corpus	88.9%
Survives RRF fusion	83.3%
Survives cross-encoder re-ranking	61.1%
Survives source-diversity cap	61.1%

Extraction loses 2 of 18 questions, and both are the intentionally unanswerable "ambiguous" probes. `temporal` scores 2/2 and `table_list` 1/1 — the exact categories the earlier analysis blamed. The 22-point drop is entirely at re-ranking.

Retriever ablation on the same questions:

Configuration	Recall
BM25 only	52.9%
Dense only	70.6%
Hybrid (RRF)	64.7%
Hybrid + cross-encoder (production)	61.1%

The full pipeline scores below its own simplest component.

6.2 The agentic pipeline silently stopped using its tools on the local backend

The most serious defect found in this project was invisible in the output.

After migrating to local models, the five-agent pipeline still ran, still cost nothing, and still produced a fluent comparative report. The execution trace showed something else: **one LLM turn per agent and zero tool spans**. The Internal Researcher never called `search_corpus`. The External Fact-Checker never called `web_search`. Every agent answered from the model's own parametric memory — precisely the behaviour a retrieval-augmented system exists to prevent — and the result was indistinguishable from a working pipeline without opening the trace.

The Visualizer made the mechanism visible. Instead of emitting a tool call it printed, as prose:

```
The JSON function call(s) that best answers the given prompt are:
{ "name": "generate_chart", "parameters": { ... } }
```

Isolated cause. A controlled probe separates prompt length from model capability:

Condition	Structured tool call emitted?
Short prompt, one tool	yes
The real agent system prompt, one tool	no — JSON printed as text
The real agent system prompt + <code>tool_choice="required"</code>	yes

`llama3.1:8b` is capable of structured tool calling and drops out of it when the system prompt is long and instruction-dense. This is a general hazard when porting agentic code from a frontier model to a small local one: the API contract is identical, the failure is silent, and the output still looks right.

Why it was not caught earlier. Every previous full-pipeline run used the hosted backend, where tool calling was reliable. On the local backend only the Gradio application and the `corpus_only` route had been exercised, both of which work. A single-tool smoke probe passed. The defect lived exactly in the gap between what was migrated and what was re-tested — and the observability layer built in §3 is what surfaced it, since no output-level check would have.

Fix. Two layers. First, `tool_choice="required"` on the opening turn of each agent forces structured output; subsequent turns return to `"auto"` so an agent can still terminate rather than being compelled to call tools forever. Second, a recovery parser extracts tool calls a model printed as text and dispatches them, restricted to the tools that agent is actually granted, so stray JSON in an answer cannot invoke anything out of role.

A third layer was required: a tool-call budget. With the first two fixes the pipeline used its tools but did not converge — the Internal Researcher issued search after search for over seventeen minutes, because each new result gave it more material to react to. Frontier models stop on their own; this one did not. After four tool calls the tools are now *withdrawn from the request entirely* and the agent is instructed to answer from what it has. Withdrawal is necessary: leaving the tools visible with `tool_choice="none"` still invites the model to narrate calls it cannot make.

Verified end to end. The complete four-agent pipeline was then re-run on the local backend:

Measure	Before fix	After fix
Tool spans in trace	0	8
Corpus searches	0	4
Web searches	0	4
Report grounded in retrieval	no	yes
Wall time	241s	1,266s
Cost	\$0.00	\$0.00

The trace now shows `internal_researcher` alternating five reasoning turns with four `search_corpus` calls, `external_fact_checker` doing the same against `web_search`, and the Synthesizer producing the report from both streams.

Two costs are worth stating plainly. The grounded pipeline is roughly five times slower than the ungrounded one, which is the true price of running a five-agent architecture on an 8B local model. And the recovery layer is load-bearing rather than defensive: two of the Fact-Checker's dispatch spans completed in 0.4 ms and 0.2 ms, which are recovered text-emitted calls rather than real round trips. Even with a forced opening turn, this model lapses back into prose mid-conversation, so removing the recovery parser would return tool use to being intermittent.

6.3 Criticism of that finding before acting on it

The cross-encoder was **not** removed, for three reasons:

1. **The gold proxy may be biased against it.** Gold is matched by lexical overlap with the reference answer. Dense retrieval optimises for roughly the same surface similarity, whereas a cross-encoder promotes passages that *answer* the question while possibly sharing less vocabulary. That would register as "the re-ranker hurts" even where it helps.
2. **n = 17 is far too small.** Four flipped questions produce the whole gap; the 95% confidence interval on 61% vs 83% at this n comfortably spans zero.
3. **The mechanism is plausible, not proven.** "A 6-layer model trained on short web passages mishandles 800-character legal text" is a hypothesis the experiment was not designed to test.

The measured alternative is **blending** rather than removal: normalise the cross-encoder score and the fusion prior to [0,1] and combine them, so the re-ranker refines the fusion ranking instead of overriding it. Blend weights of 0.3–0.5 recovered 70.6% while keeping the re-ranker in place. This is offered as a recommendation, not applied silently.

6.4 A superseded analysis, retained for the record

This analysis is incorrect. It attributes the retrieval shortfall to table and date extraction. A stage-by-stage attribution (§7.2) later showed extraction loses only 2 of 18 questions — both intentionally unanswerable — while the `temporal` category scores 2/2 and `table_list` 1/1. The real loss is at cross-encoder re-ranking. The reasoning below is preserved to show how a plausible, internally-consistent diagnosis can be wrong when the metric conflates several pipeline stages into a single number.

The two evaluation methods agree on the diagnosis. Seven of nine judged failures are classified `retrieval_miss`, and RAGAS context precision (0.432) is the weakest metric — the retriever is surfacing passages that are related but not the ones containing the answer. Faithfulness at 0.60 exceeding context precision at 0.43 indicates the generator is behaving reasonably given what it receives: it is not inventing much beyond its context, it is being handed the wrong context.

The category breakdown localises this further. Perfect scores on `contradictory`, `edge_of_chunk`, and `multi_part` show the chunking strategy and the overlap policy work as intended. The zeros cluster on categories that stress lexical matching:

- **`table_list` and `temporal` failures** trace to numeric and tabular content. Question q01 required two specific dates (14 June 2027 transposition, 14 June 2032 evaluation); q08 required comparing median survey durations across two countries. Dates and table cells survive extraction as text but carry almost no

distinctive embedding signal, and BM25 treats them as rare tokens that a paraphrased query never contains.

- **negation and paraphrased failures** are the classic dense-retrieval weakness: "does the toolkit include X" and "what is *not* covered" produce embeddings close to passages asserting the opposite.
- **multi_hop** retrieved correctly but generated incorrectly — the one case where the failure is genuinely downstream.

This is consistent with the principle that the quality of the data pipeline bounds the quality of the agent. Our remedies would be data-engineering remedies: table-aware extraction preserving cell structure, and metadata enrichment extracting dates and article numbers into a filterable field rather than leaving them as undifferentiated text.

We report these numbers as they are. A 53% pass rate on a benchmark that is 73% hard questions is a weak absolute result and we do not present it otherwise; its value is diagnostic. Cell counts of one to two questions per category mean the per-category figures indicate direction, not effect size.

4.4 Observability findings

Tracing surfaced a behaviour invisible from the output alone: in the demonstration run the Synthesizer called `create_markdown_report` four times in a single turn, writing four near-identical files. The final report was correct, so no output-level check would have caught it — but the waterfall made the redundancy immediately visible. This is the concrete argument for tracing: correctness of the answer does not imply efficiency of the process.

Building the tracer also revealed a measurement artefact worth documenting. Because the agent dispatcher and the MCP tool each open a span for one logical call, naive counting double-counted every tool invocation (reporting 26 instead of 11). We resolved it by counting only tool spans with no tool ancestor, and added a regression test. We note it because it is the kind of error that makes an observability dashboard confidently wrong.

7. Discussion

7.1 What the experiments actually show

The binding constraint on this system is retrieval ranking, not generation and not extraction. Generation faithfulness (0.60) exceeds context precision (0.43), which says the generator behaves reasonably given what it receives and is being handed the wrong context. Extraction loses 2 of 18 questions, both intentionally unanswerable. The 22-point loss sits at re-ranking.

Two results are worth more than the headline numbers:

The chunk-size effect reverses by retriever. Dense degrades at 1200 characters while BM25 improves. A system running both must choose a size that penalises neither, and a study reporting one aggregate number would have picked a size that quietly damages one half of its own retriever.

Overlap is invisible to recall and visible to MRR. It changes no recall figure under any retriever and improves ranking under all three. An evaluation reporting recall alone would have deleted it to save 9% of index size and

degraded every downstream answer by burying the right passage deeper in the context window.

7.2 Methodological errors caught in our own analysis

The embedding comparison was meaningless in its first form. It reported all three models at byte-identical recall, MRR, and precision. Three different neural models cannot do that. The cause was in the reporting code, not the pipeline: the aggregation took the **best cell** per level, and the best cell for every embedder was the **BM25** row — and BM25 never touches an embedding, so those rows are identical by construction. The table was comparing three copies of the same lexical retriever and labelling it an embedding comparison. The fix holds confounders fixed when marginalising over a factor.

The same bug survived into the chunk-size table and was caught only when a plotted figure disagreed with it: the figure held the retriever fixed and showed 1200 characters *losing* 6.7 points, while the table showed it tying. Both numbers were real; the table's aggregation was not answering the question it claimed to. Correcting it produced the retriever interaction reported in §7.4, which is a better result than the one the bug concealed.

We record both because they are exactly the kind of table that is confidently wrong and would survive review: the numbers were real, the comparison was not.

7.3 Limitations

- **n = 15–18.** Every difference reported here is within sampling noise. These results rank configurations weakly; they do not establish effect sizes. `scripts/build_ground_truth.py` generates a larger silver-standard set locally to address this, deliberately kept **separate** from the human set because questions generated *from* a chunk make that chunk artificially easy to retrieve, inflating absolute recall.
- **The factorial ran on a 1,723-page subset**, not the full 2,247-page corpus, so absolute numbers are not comparable to §4.2. Relative comparisons are valid because the subset is held constant across conditions.
- **End-to-end judge and RAGAS figures in §4.2 are hosted-backend measurements**, taken before the migration, and are labelled as such. The harnesses themselves are now fully local, so re-running reproduces them on the local stack; the re-index that feeds them is the long pole at ~14–20 chunks/s for 12,863 chunks.
- **A real defect surfaced during that re-index.** The first local build aborted at ~7,300 chunks with a 400 from the embedding endpoint: `nomic-embed-text` has a 2,048-token context and 11 chunks — stitched pages and table-derived text, made longer by the metadata header — exceed it. The unguarded request discarded 7,300 chunks of completed work over 11 bad inputs. The fix truncates inputs to a safe budget and falls back to per-item embedding if a batch still fails. A local backend has tighter limits than a hosted one, and a pipeline ported across backends inherits none of the old provider's tolerance.

7.4 Work in progress at time of writing

Stated plainly rather than omitted:

- **The agentic pipeline is verified on the local backend but is slow.** A complete four-agent run takes about 21 minutes and depends on a recovery layer that repairs malformed tool calls (§6.2). It is functional,

not comfortable. Latency figures quoted elsewhere for the agentic path derive from hosted-backend runs and are roughly five times optimistic for local execution.

These are gaps in evidence, not in implementation: every harness is present, tested and documented in the README.

8. Conclusion

We built an agentic RAG system for EU women's safety policy in which the agentic machinery is deliberately constrained rather than maximised. The five agents are narrow, their tool access is bounded in code, every untrusted input is scanned including material retrieved from our own store, and no output reaches a person without a human approving it against a displayed cost and security summary.

The honest headline is that retrieval, not agency, is our binding constraint. Generation faithfulness (0.60) exceeds context precision (0.43), and seven of nine failures are retrieval misses concentrated in tabular, temporal, and negated queries. The agent team, the security layer, and the observability stack all function as designed; they are built on a retriever that needs table-aware extraction and metadata filtering to reach the accuracy this domain warrants. That is a data-engineering problem, which is where we would direct the next iteration.

The security and observability layers earned their place during development rather than after it. The observability layer caught a redundant tool-call pattern no output check would have found. The security test suite caught a deadlock in our own guard — a non-reentrant lock acquired twice on the rate-limiting path — that would have hung the system under concurrent load. Both are arguments for building these layers as first-class components rather than as instrumentation bolted on at the end.

Appendix A — Repository structure

app/	
mcp_server.py	MCP server: 8 tools (Steps 1–4 + extensions)
agents.py	5-agent team, role permissions, tool schemas
main.py	Supervisor router + HITL review gate (Step 3)
app.py	Gradio web interface
teams_bot.py	Microsoft Teams webhook (Step 5)
security.py	Prompt injection, leakage, output validation
observability.py	Hierarchical tracing, token/cost, context capture
scripts/	
extract.py	PDF extraction (PyMuPDF + pdfplumber fallback)
chunk.py	Sentence-aware chunking, embedding, hybrid retrieval
eval.py	LLM-as-judge scorecard
ragas_eval.py	RAGAS faithfulness / precision / recall
cost_function.py	Persistent cost ledger with budget enforcement
retrieval_check.py	Retrieval diagnostics
tests/	207 offline tests
reports/	
approved/	Human-approved published reports
charts/	matplotlib output
diagrams/	Mermaid output
traces/	Per-request execution traces

Appendix B — Requirement coverage

Build steps

Requirement	Implementation
Step 1 — MCP server with live web search	<code>web_search</code> , <code>scrape_url</code>
Step 2 — Multi-agent team	5 agents in <code>app/agents.py</code>
Step 2 — <code>create_markdown_report</code> tool	<code>app/mcp_server.py</code>
Step 3 — Human-in-the-loop	<code>hitl_review()</code> in <code>app/main.py</code>
Step 3 — Supervisor / router agent	<code>supervisor_route()</code>
Step 4 — <code>add_to_database</code> tool	<code>app/mcp_server.py</code>
Step 4 — PDF ingestion via MCP	<code>ingest_pdf</code>
Step 5 — Messaging integration	Teams webhook + Slack publisher
Extension — Visualizer agent	<code>agent_visualizer</code>
Extension — matplotlib charts	<code>generate_chart</code>
Extension — Mermaid diagrams	<code>create_mermaid_diagram</code>

Corpus processing

Requirement	Where	§
Text extraction	<code>scripts/extract.py</code> — PyMuPDF + pdfplumber fallback + table-aware	8.1
Data cleaning	<code>clean_text</code> , <code>detect_repeated_lines</code>	8.2
Document segmentation	<code>_split</code> — sentence-aware, overlapping	8.3
Metadata management	chunk provenance + <code>scripts/metadata.py</code> identifiers	8.4
Impact of chunk size	measured, 400 / 800 / 1200	7.4, 8.5
Impact of chunk overlap	measured, 0 / 100 / 200	7.4, 8.5
Structure preservation	measured, metadata headers on/off	7.4, 8.5

Configuration comparison

Requirement	Where	\$
Different embedding models	MiniLM-L6 / bge-small / e5-small, dense held fixed	7.4
Different chunk sizes	400 / 800 / 1200	7.4
Different k	3 / 5 / 8 / 12 / 20, per retriever	7.4
Different generation models	<code>LLM_BACKEND</code> + <code>OLLAMA_MODEL</code> select any local model	7.1
Rigour over score-chasing	failure-stage attribution, self-criticism of the reranker finding, corrected aggregation bug, stated limitations	7.2–7.6

Security, testing, observability

Requirement	Implementation
Security	<code>app/security.py</code> + 64 tests
Testing — LLM-as-judge	<code>scripts/eval.py</code> (local judge)
Testing — RAGAS	<code>scripts/ragas_eval.py</code> (local judge + embeddings)
Testing — unit	249 offline tests
Observability — tracing	<code>app/observability.py</code>
Observability — token & cost	<code>scripts/cost_function.py</code> + tracer
Observability — context debugging	prompt-window capture per LLM span